

Optimization-based Trajectory Planning with Strictly Safe Collision Avoidance Between Adjacent Collocation Points

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Abstract—.

Index Terms—Computational optimal control, non-collocation point error, collision avoidance, trajectory planning

I. INTRODUCTION

II. RELATED WORK

III. PROBLEM STATEMENT

This section states the discretization-induced safety gap and introduces the buffered-footprint idea used to close it.

A. Nominal OCP Formulation and Its Limitation

Consider a trajectory planning problem over a finite time horizon $[0, T]$. Let $\mathbf{z}(t) \in \mathbb{R}^{n_z}$ denote the state vector of the ego vehicle and $\mathbf{u}(t) \in \mathbb{R}^{n_u}$ denote the control input. A nominal optimization-based trajectory planner can be formulated as

$$\begin{aligned} \min_{\mathbf{z}(t), \mathbf{u}(t), T} \quad & J(\mathbf{z}(t), \mathbf{u}(t), T) \\ \text{s.t.} \quad & \dot{\mathbf{z}}(t) = \mathbf{f}(\mathbf{z}(t), \mathbf{u}(t)), t \in [0, T]; \\ & \underline{\mathbf{z}} \leq \mathbf{z}(t) \leq \bar{\mathbf{z}}, \underline{\mathbf{u}} \leq \mathbf{u}(t) \leq \bar{\mathbf{u}}, t \in [0, T]; \\ & \mathbf{z}(0) = \mathbf{z}_{\text{init}}, \mathbf{u}(0) = \mathbf{u}_{\text{init}}; \\ & \mathbf{z}(T) = \mathbf{z}_{\text{end}}, \mathbf{u}(T) = \mathbf{u}_{\text{end}}; \\ & fp(\mathbf{z}(t)) \subseteq Y_{\text{FREE}}, t \in [0, T]. \end{aligned} \quad (1)$$

where J is the objective function, $\mathbf{f}$ describes the vehicle kinematics or dynamics, and $\bar{\mathbf{z}}, \underline{\mathbf{z}}, \bar{\mathbf{u}}, \underline{\mathbf{u}}$ define the admissible state and control bounds. The mapping $fp(\mathbf{z}(t)) \subset \mathbb{R}^2$ denotes the physical footprint occupied by the vehicle at state $\mathbf{z}(t)$, and denotes the free space.

Since (1) is a continuous-time OCP, it is commonly transcribed into a finite-dimensional nonlinear programming

(NLP) problem. Let the time horizon be discretized by N_{fe} collocation points,

$$0 = t_1 < t_2 < \dots < t_{N_{\text{fe}}} = T, \quad (2)$$

with $\Delta t_k = t_{k+1} - t_k, k = 1, \dots, N_{\text{fe}} - 1$. Denote $\mathbf{z}_k = \mathbf{z}(t_k)$ and $\mathbf{u}_k = \mathbf{u}(t_k)$. A direct transcription scheme yields an NLP of the form

$$\begin{aligned} \min_{\mathbf{z}_k, \mathbf{u}_k, \Delta t_k} \quad & J_d(\mathbf{z}_k, \mathbf{u}_k, \Delta t_k) \\ \text{s.t.} \quad & \mathbf{z}_{k+1} = \Phi(\mathbf{z}_k, \mathbf{u}_k, \Delta t_k), k = 1, \dots, N_{\text{fe}} - 1; \\ & \mathbf{G}_d(\mathbf{z}_k, \mathbf{u}_k) \leq 0, k = 1, \dots, N_{\text{fe}} - 1; \end{aligned} \quad (3)$$

where Φ is the discrete transition map and $\mathbf{G}_d$ collects the discretized state, control, boundary, and collision-avoidance constraints.

The limitation of this transcription is that the continuous collision-avoidance condition in (1) is reduced to a finite set of checks at the collocation points. Thus, $fp(\mathbf{z}(t)) \subseteq Y_{\text{FREE}}$ for all k does not necessarily imply that the vehicle is collision-free during the continuous motion from t_k to t_{k+1} . As illustrated in Fig. 1, the swept region between two adjacent collocation points may still intersect obstacles even when both endpoint footprints are safe. Therefore, a nominal discretized OCP lacks a direct mechanism for guaranteeing collision avoidance between adjacent collocation points.

B. Fundamental Idea of Safety Guarantees Between Adjacent Collocation Points

To address this limitation, this paper replaces the nominal footprint at each collocation point with a buffered footprint. The buffered footprint is an enlarged rectangle constructed from the nominal vehicle footprint at t_k . Its role is to cover the entire region swept by the vehicle while moving from t_k to t_{k+1} . Once this coverage is guaranteed, imposing collision avoidance on the buffered footprint at t_k is sufficient to guarantee collision avoidance throughout the corresponding inter-collocation interval.

Specifically, let $A_k B_k C_k D_k$ denote the nominal rectangular vehicle footprint at the k -th collocation point. As shown in Fig. 2, a buffered rectangle $\hat{A}_k \hat{B}_k \hat{C}_k \hat{D}_k$ is constructed by expanding $A_k B_k C_k D_k$ along the four directions of the vehicle-fixed frame. The four expansion distances are denoted by $e_{\text{forward}}(t_k)$, $e_{\text{back}}(t_k)$, $e_{\text{right}}(t_k)$, and $e_{\text{left}}(t_k)$. Here, $e_{\text{forward}}(t_k)$ and $e_{\text{back}}(t_k)$ are the longitudinal buffering variables in the forward and rearward directions, while $e_{\text{left}}(t_k)$ and $e_{\text{right}}(t_k)$ are the lateral buffering variables. The central requirement is that the buffered rectangle $\hat{A}_k \hat{B}_k \hat{C}_k \hat{D}_k$ fully contains the swept region generated by the nominal footprint from t_k to t_{k+1} .

Therefore, the remaining problem is to determine these four buffering variables for each adjacent pair of collocation points.

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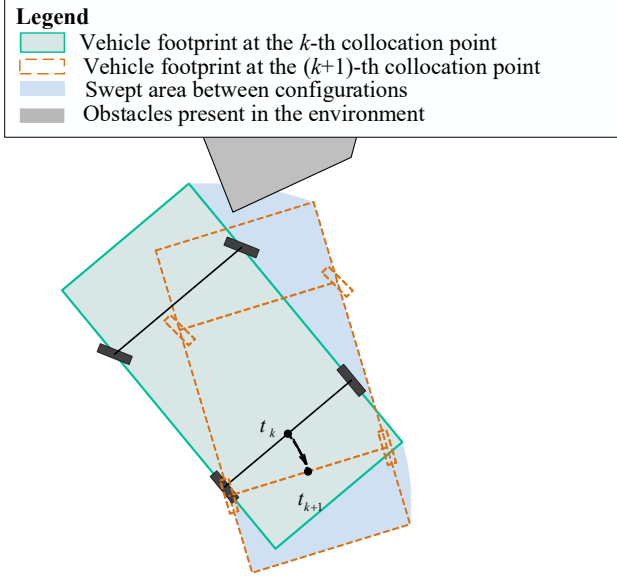
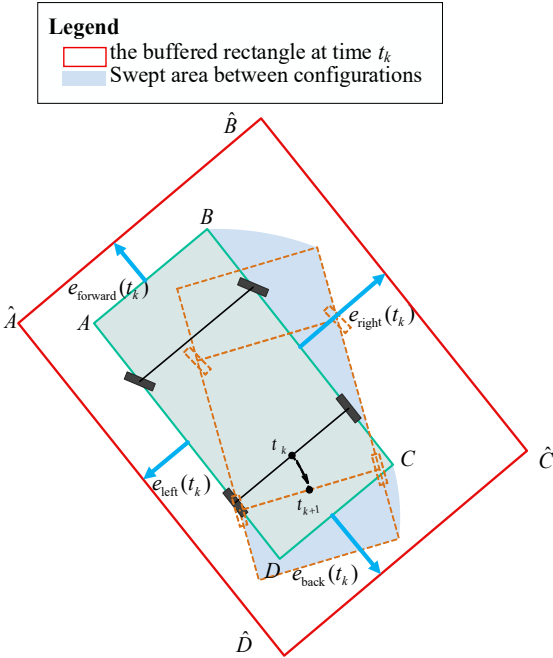


Fig. 1. Collision risk between adjacent collocation points


 Fig. 2. Buffered footprint $\hat{A}_k \hat{B}_k \hat{C}_k \hat{D}_k$ at t_k

Section IV derives their unified expressions for both forward and reverse driving and formulates the resulting buffered-footprint constraints for NLP-based trajectory planning.

IV. SAFE COLLISION-AVOIDANCE CONSTRAINTS

This section derives the buffered-footprint constraints introduced in Section III. The key task is to determine the four buffering variables at each collocation point so that the buffered footprint covers the continuous swept region over the next inter-collocation interval. The derivation is based on the local motion segment induced by the discretized trajectory.

A. Definitions of Buffering Variables

At the k -th collocation point, the buffered footprint constructed at t_k must cover the region swept by the vehicle from t_k to t_{k+1} . Since the vehicle velocity and steering angle may vary across collocation points, the swept region is different for each interval. Therefore, the four buffering variables should be expressed as functions of the local motion state at t_k .

Let $\phi_k = \phi(t_k)$ denote the steering angle, $v_k = v(t_k)$ denote the longitudinal velocity, and $\Delta t_k = t_{k+1} - t_k$ denote the interval length defined in Section III. The signed curvature and signed travel distance over the interval $[t_k, t_{k+1}]$ are defined as

$$\kappa_k := \frac{\tan \phi_k}{L_w}, s_k := \Delta t_k \cdot v_k, \quad (4)$$

where L_w is the wheelbase. To handle forward and reverse driving in a unified manner, the signed distance s_k is decomposed into its forward and reverse components,

$$s_k^+ := \max(s_k, 0), s_k^- := \max(-s_k, 0). \quad (5)$$

Here, s_k^+ is the forward travel distance, and s_k^- is the reverse travel distance. At most one of them is nonzero for a given interval.

Based on these auxiliary variables, the four buffering variables at t_k are defined as

$$\begin{aligned} e_{\text{forward}}(t_k) &= s_k^+ \cdot \left(1 + \frac{L_B}{2} \cdot |\kappa_k|\right), \\ e_{\text{back}}(t_k) &= s_k^- \cdot \left(1 + \frac{L_B}{2} \cdot |\kappa_k|\right), \\ e_{\text{left}}(t_k) &= \max \left\{ -L_R \cdot \kappa_k \cdot s_k^+, \left(L_F + \frac{s_k^+}{2}\right) \cdot \kappa_k \cdot s_k^+ \right\} \\ &\quad + \max \left\{ -L_F \cdot \kappa_k \cdot s_k^-, \left(L_R + \frac{s_k^-}{2}\right) \cdot \kappa_k \cdot s_k^- \right\}, \\ e_{\text{right}}(t_k) &= \max \left\{ L_R \cdot \kappa_k \cdot s_k^+, -\left(L_F + \frac{s_k^+}{2}\right) \cdot \kappa_k \cdot s_k^+ \right\} \\ &\quad + \max \left\{ L_F \cdot \kappa_k \cdot s_k^-, -\left(L_R + \frac{s_k^-}{2}\right) \cdot \kappa_k \cdot s_k^- \right\}, \end{aligned} \quad (6)$$

In these expressions, L_F is the distance from the rear-axle reference point to the front boundary of the vehicle, L_R is the distance from the rear-axle reference point to the rear boundary, and L_B is the vehicle width. The longitudinal buffers $e_{\text{forward}}(t_k)$ and $e_{\text{back}}(t_k)$ cover the forward and rearward sweep of the vehicle, respectively. The lateral buffers $e_{\text{left}}(t_k)$ and $e_{\text{right}}(t_k)$ account for the sideward sweep induced by the local turning motion. The use of s_k^+ and s_k^- allows the same expressions to cover both forward and reverse maneuvers.

The validity of these buffering variables relies on three conditions. First, the local motion over each interval is approximated by a constant-speed, constant-curvature segment. This is consistent with a direct-transcription formulation in which v_k and ϕ_k define the local motion from t_k to t_{k+1} . Under this assumption, the vehicle moves along a circular arc when $\kappa_k \neq 0$ and along a straight segment when $\kappa_k = 0$.

Second, the vehicle geometry and steering bound should satisfy

$$2L_w > L_B \cdot \tan \phi_{\max}, \quad (7)$$

where $\phi_{\max}$ is the maximum admissible steering angle. This condition ensures that the curvature-induced longitudinal

buffering terms remain well defined under the steering limits.

Third, the local motion should satisfy the following prerequisite constraints:

$$\begin{aligned}
|\kappa_k \cdot s_k| &\leq \frac{\pi}{2}, \\
\left(1 + \frac{L_B}{2} \cdot |\kappa_k|\right) \cdot \tan(|\kappa_k| \cdot s_k^+) &\leq L_R \cdot |\kappa_k|, \\
|\kappa_k| \cdot L_F \cdot \tan(|\kappa_k| \cdot s_k^+) &\leq 1 + \frac{L_B}{2} \cdot |\kappa_k|, \\
\left(1 + \frac{L_B}{2} \cdot |\kappa_k|\right) \cdot \tan(|\kappa_k| \cdot s_k^-) &\leq L_F \cdot |\kappa_k|, \\
|\kappa_k| \cdot L_R \cdot \tan(|\kappa_k| \cdot s_k^-) &\leq 1 + \frac{L_B}{2} \cdot |\kappa_k|.
\end{aligned} \tag{8}$$

These constraints restrict the local arc length and turning angle so that the buffered rectangle constructed at t_k can serve as a valid envelope of the swept vehicle footprint over $[t_k, t_{k+1}]$. Together, (6)-(8) define the buffering variables and their applicable range. The detailed geometric derivation and proof are provided in the Appendix.

B. Buffered-Footprint Collision Constraints and Strictly Safe NLP Formulation

In Section IV-A, (4)-(6) define the local curvature, interval displacement, and four buffering variables, while (7)-(8) specify the geometric conditions under which these variables are valid. This subsection embeds the resulting buffered footprint into the collision constraints and formulates the strictly safe NLP.

At the k -th collocation point, let (x_k, y_k) be the rear-axle reference point and θ_k the heading angle. Using the buffering variables from (6), the buffered footprint is an enlarged rectangle whose forward, rearward, leftward, and rightward extensions are

$$L_F + e_{\text{forward},k}, L_R + e_{\text{back},k}, \frac{L_B}{2} + e_{\text{left},k}, \frac{L_B}{2} + e_{\text{right},k} \tag{9}$$

Thus, its four vertices are

$$\begin{aligned}
\hat{A}_k &= \begin{bmatrix} x_k \\ y_k \end{bmatrix} + (L_F + e_{\text{forward},k}) \cdot \begin{bmatrix} \cos \theta_k \\ \sin \theta_k \end{bmatrix} \\
&\quad + \left(\frac{L_B}{2} + e_{\text{left},k}\right) \cdot \begin{bmatrix} -\sin \theta_k \\ \cos \theta_k \end{bmatrix}, \\
\hat{B}_k &= \begin{bmatrix} x_k \\ y_k \end{bmatrix} + (L_F + e_{\text{forward},k}) \cdot \begin{bmatrix} \cos \theta_k \\ \sin \theta_k \end{bmatrix} \\
&\quad - \left(\frac{L_B}{2} + e_{\text{right},k}\right) \cdot \begin{bmatrix} -\sin \theta_k \\ \cos \theta_k \end{bmatrix}, \\
\hat{C}_k &= \begin{bmatrix} x_k \\ y_k \end{bmatrix} - (L_R + e_{\text{back},k}) \cdot \begin{bmatrix} \cos \theta_k \\ \sin \theta_k \end{bmatrix} \\
&\quad - \left(\frac{L_B}{2} + e_{\text{right},k}\right) \cdot \begin{bmatrix} -\sin \theta_k \\ \cos \theta_k \end{bmatrix}, \\
\hat{D}_k &= \begin{bmatrix} x_k \\ y_k \end{bmatrix} - (L_R + e_{\text{back},k}) \cdot \begin{bmatrix} \cos \theta_k \\ \sin \theta_k \end{bmatrix} \\
&\quad + \left(\frac{L_B}{2} + e_{\text{left},k}\right) \cdot \begin{bmatrix} -\sin \theta_k \\ \cos \theta_k \end{bmatrix}.
\end{aligned} \tag{10}$$

Let

$$\mathcal{B}_k = \text{poly}(\hat{A}_k, \hat{B}_k, \hat{C}_k, \hat{D}_k) \tag{11}$$

denote the buffered footprint at the k -th collocation point. Under the validity conditions in (7)-(8), $\mathcal{B}_k$ covers the continuous swept footprint from t_k to t_{k+1} . Therefore, enforcing obstacle avoidance for $\mathcal{B}_k$ at the collocation point provides continuous safety over the corresponding interval.

Obstacles are not restricted to rectangles. So let the j -th obstacle be a convex polygon,

$$\mathcal{O}_j = \text{poly}(P_{j,1}, P_{j,2}, \dots, P_{j,M_j}), \quad j = 1, \dots, N_{\text{obs}}, \tag{12}$$

where M_j is its number of vertices. To impose a practical algebraic exclusion condition associated with

$$\mathcal{B}_k \cap \mathcal{O}_j = \emptyset \tag{13}$$

we employ a triangle-area-based vertex-outside test. For a point p and a convex polygon

$$\mathcal{Q} = \text{poly}(Q_1, Q_2, \dots, Q_M), \tag{14}$$

whose vertices are ordered consecutively along its boundary, define

$$\Psi(p, \mathcal{Q}) = \sum_{m=1}^M \frac{1}{2} |(Q_m - p) \times (Q_{m+1} - p)|, \quad Q_{M+1} = Q_1. \tag{15}$$

If p lies inside or on the boundary of $\mathcal{Q}$, then $\Psi(p, \mathcal{Q}) = \text{Area}(\mathcal{Q})$; otherwise, $\Psi(p, \mathcal{Q})$ is larger. Hence, the outside condition is imposed by

$$\Psi(p, \mathcal{Q}) \geq \text{Area}(\mathcal{Q}) + \epsilon \tag{16}$$

where $\epsilon > 0$ is a small positive tolerance used to exclude boundary contact in the area test. The collision constraints are imposed in both directions. First, all obstacle vertices must lie outside the buffered footprint:

$$\Psi(P_{j,m}, \mathcal{B}_k) \geq \text{Area}(\mathcal{B}_k) + \epsilon, \quad m = 1, \dots, M_j. \tag{17}$$

Second, the four buffered-footprint vertices must lie outside the obstacle polygon:

$$\begin{aligned}
\Psi(A_k, \mathcal{O}_j) &\geq \text{Area}(\mathcal{O}_j) + \epsilon, \\
\Psi(B_k, \mathcal{O}_j) &\geq \text{Area}(\mathcal{O}_j) + \epsilon, \\
\Psi(C_k, \mathcal{O}_j) &\geq \text{Area}(\mathcal{O}_j) + \epsilon, \\
\Psi(D_k, \mathcal{O}_j) &\geq \text{Area}(\mathcal{O}_j) + \epsilon.
\end{aligned} \tag{18}$$

Constraints (17)-(18) are imposed at collocation points $k = 2, \dots, N_{\text{fc}} - 1$ for all obstacles $j = 1, \dots, N_{\text{obs}}$. The initial point $k = 1$ is excluded because the prescribed boundary conditions impose zero initial velocity, acceleration, steering angle, and steering rate. Consequently, under the adopted discrete vehicle model, the first inter-collocation interval is stationary. Since the prescribed initial configuration is collision-free, no additional buffered-footprint collision constraint is required at $k = 1$. The terminal point $k = N_{\text{fr}}$ requires no buffered-footprint constraint because no subsequent inter-collocation interval originates from it.

This formulation extends the collision model to general convex polygonal obstacles, while a nonconvex obstacle can be represented through convex decomposition. The area test is not a complete separating-axis test for polygon intersection; rather, it provides a practical bidirectional vertex-exclusion condition. With a collision-free initial trajectory and continuous optimization without topological crossing, it prevents collision configurations from emerging through vertex penetration.

Combined with the buffered-footprint construction, this condition is used to enforce collision avoidance over the inter-collocation intervals.

Combining the buffered-footprint construction and the collision constraints, we formulate the resulting NLP with a quadratic time-step objective. This objective favors shorter traversal duration while discouraging excessively nonuniform temporal discretization; for a fixed total duration, it is minimized when all time intervals are equal. The resulting NLP is

$$\begin{aligned}
\min_{\mathcal{Z}} J_{\Delta t} &:= \sum_{k=1}^{N_{fe}-1} (\Delta t_k)^2 \\
\text{s.t. } T &= \sum_{k=1}^{N_{fe}-1} \Delta t_k, T \leq T_{\max}, \\
\Delta t_{\min} &\leq \Delta t_k \leq \Delta t_{\max}, k=1, \dots, N_{fe}-1, \\
\mathbf{x}_{k+1} &= \mathbf{x}_k + \Delta t_k \cdot \begin{bmatrix} v_k \cos \theta_k \\ v_k \sin \theta_k \\ v_k \tan \phi_k / L_W \\ a_k \\ \omega_k \end{bmatrix}, k=1, \dots, N_{fe}-1, \\
\mathbf{x}_1 &= \mathbf{x}_{\text{init}}, \quad \mathbf{x}_{N_{fe}} = \mathbf{x}_{\text{goal}}, \\
\mathbf{u}_1 &= \mathbf{u}_{\text{init}}, \quad \mathbf{u}_{N_{fe}} = \mathbf{u}_{\text{goal}}, \\
\begin{bmatrix} a_{\min} \\ v_{\min} \\ -\Omega_{\max} \\ -\Phi_{\max} \end{bmatrix} &\leq \begin{bmatrix} a_k \\ v_k \\ \omega_k \\ \phi_k \end{bmatrix} \leq \begin{bmatrix} a_{\max} \\ v_{\max} \\ \Omega_{\max} \\ \Phi_{\max} \end{bmatrix}, \\
&\text{motion definitions in (4)-(5),} \\
&\text{buffering-variable equations in (6),} \\
&\text{buffering validity conditions in (7)-(8),} \\
&\text{buffered-footprint vertex construction in (10),} \\
&\text{triangle-area collision constraints in (17)-(18).}
\end{aligned} \tag{19}$$

where

$$\mathbf{x}_k = [x_k, y_k, \theta_k, v_k, \phi_k]^T, \mathbf{u}_k = [a_k, \omega_k]^T \tag{20}$$

The decision variable set is

$$\mathcal{Z} = \left\{ \mathbf{x}_k, \mathbf{u}_k, \Delta t_k, T, \kappa_k^+, s_k^+, s_k^-, e_{\text{forward},k}, e_{\text{back},k}, e_{\text{left},k}, e_{\text{right},k}, \hat{A}_k^x, \hat{A}_k^y, \hat{B}_k^x, \hat{B}_k^y, \hat{C}_k^x, \hat{C}_k^y, \hat{D}_k^x, \hat{D}_k^y \right\} \tag{21}$$

Here, Δt_k is a decision variable, not a fixed sampling period. The collocation points are therefore nonuniform in time. The optimizer jointly adjusts the states, controls, time intervals, and total time. Meanwhile, the validity conditions in (7)-(8) are included directly in the NLP feasible set, ensuring that the buffered footprint in (6) remains geometrically valid throughout optimization. In this way, the finite-point safety gap identified in Section III is converted into a directly solvable strictly safe NLP.

V. NUMERICAL SOLUTION AND SOLVER-AIDING STRATEGIES

The previous section formulated the strictly safe trajectory optimization problem as a nonconvex NLP. This section further addresses its numerical solution and associated solver-aiding

strategies. A hybrid A* reference trajectory is used to construct the initial solution, which is then optimized using an interior-point method. To address nonsmooth max operators, collisions of the initial buffered footprints, and inappropriate initialization of the variable time intervals, LSE smoothing, geometric feasibility enhancement, and adaptive temporal discretization are introduced, respectively.

A. Variables Interior-Point Solution and Constraint Smoothing

The nonconvex NLP in (19) is solved using an interior-point method. The interior-point method solves a sequence of barrier subproblems while progressively reducing the barrier parameter, thereby driving the iterates toward a local KKT point of the original problem. For the nonconvex trajectory optimization problem considered in this paper, the structured initial solution constructed from hybrid A* provides the optimizer with a reasonable motion topology and state-control profile, thereby facilitating the subsequent local search.

Nevertheless, the buffered-footprint model still contains nonsmooth structures induced by max operators. Equation (5) uses max operators to decompose the signed travel distance into forward and reverse components, while the leftward and rightward buffering variables in (6) each contain two additional max operators. When two candidate branches attain the same value, the active branch switches and the classical derivative of the max operator is undefined. To improve the numerical behavior of these expressions in gradient-based NLP optimization, the log-sum-exp (LSE) function is employed to provide smooth approximations of the relevant max operators.

For any two real numbers a and b , define

$$\mathcal{L}_\alpha(a, b) := \frac{1}{\alpha} \ln(e^{\alpha a} + e^{\alpha b}), \quad \alpha > 0, \tag{22}$$

where α is the smoothing parameter. The LSE function is a smooth upper approximation of the max operator and satisfies

$$0 \leq \mathcal{L}_\alpha(a, b) - \max(a, b) \leq \frac{\ln 2}{\alpha}. \tag{23}$$

Hence, as α increases, the LSE approximation approaches the original max operator. The proof of (23), together with the complete error analysis for the two-stage smoothing introduced below, is provided in the Appendix.

First, the forward and reverse travel distances in (5) are smoothed as

$$\tilde{s}_k^+ = \frac{1}{\alpha} \ln(1 + e^{\alpha s_k^+}), \quad \tilde{s}_k^- = \frac{1}{\alpha} \ln(1 + e^{\alpha s_k^-}). \tag{24}$$

In the subsequent buffered-footprint calculations, $\tilde{s}_k^+$ and $\tilde{s}_k^-$ replace s_k^+ and s_k^- , respectively, in (6). For the forward and rearward buffering variables, only this substitution is required. For the leftward and rightward buffering variables, a second-stage LSE approximation is further applied to the branch-selection max operators in (6). Accordingly, the smoothed leftward buffering variable is written as

$$\begin{aligned}
\tilde{e}_{\text{left},k} &= \mathcal{L}_\alpha \left(-L_R \kappa_k \tilde{s}_k^+, \left(L_F + \frac{s_k^+}{2} \right) \kappa_k \tilde{s}_k^+ \right) \\
&\quad + \mathcal{L}_\alpha \left(-L_F \kappa_k \tilde{s}_k^-, \left(L_R + \frac{s_k^-}{2} \right) \kappa_k \tilde{s}_k^- \right)
\end{aligned} \tag{25}$$

and the smoothed rightward buffering variable is

$$\begin{aligned} \tilde{e}_{\text{right},k} = & \mathcal{L}_\alpha \left(L_R \kappa_k \tilde{s}_k^+, - \left(L_F + \frac{s_k^+}{2} \right) \kappa_k \tilde{s}_k^+ \right) \\ & + \mathcal{L}_\alpha \left(L_F \kappa_k \tilde{s}_k^-, - \left(L_R + \frac{s_k^-}{2} \right) \kappa_k \tilde{s}_k^- \right). \end{aligned} \quad (26)$$

The resulting formulation therefore involves two stages of smoothing. The first acts on s_k^+ and s_k^- , whereas the second acts on the branch-selection max operators in $e_{\text{left},k}$ and $e_{\text{right},k}$. Consequently, the total approximation errors in the lateral buffering variables include not only the LSE approximation errors of the two outer max operators, but also the errors propagated through $\tilde{s}_k^+$ and $\tilde{s}_k^-$ from the first-stage smoothing.

As proved in the Appendix, the two-stage LSE treatment does not underestimate the original buffering variables in (6), i.e.,

$$\tilde{e}_{\text{left},k} \geq e_{\text{left},k}, \tilde{e}_{\text{right},k} \geq e_{\text{right},k}$$

and, as $\alpha \rightarrow +\infty$

$$\tilde{e}_{\text{left},k} \rightarrow e_{\text{left},k}, \tilde{e}_{\text{right},k} \rightarrow e_{\text{right},k}$$

The smoothing is therefore one-sided conservative with respect to the safety envelope: it does not reduce the theoretically required buffered footprint, but only introduces an additional expansion that vanishes as α increases.

B. Geometric Feasibility Enhancement of the Initial Solution

Hybrid A* is used in this work to provide a reference path with a reasonable motion topology, and its collision checking is performed on the physical vehicle footprint. Consequently, the hybrid A* path guarantees that the physical vehicle body does not collide with obstacles. This property, however, does not directly extend to the buffered footprint constructed in (6). Because the buffered footprint is obtained by expanding the physical footprint according to the local travel distance and curvature, an initial buffered footprint may still penetrate an obstacle even when the physical vehicle remains collision-free. This issue is particularly pronounced in narrow passages, during relatively large local steering motions, or over intervals with relatively large travel distances.

Such initial geometric infeasibility can increase the difficulty of solving the NLP. The collision-avoidance constraints in (17)–(18) exhibit strong geometric nonlinearity. If an initial buffered footprint already overlaps an obstacle, the optimizer must recover the corresponding footprint variables from a colliding configuration while simultaneously adjusting the vehicle states, controls, and time variables. Starting from a geometric configuration that remains outside the obstacles is therefore generally more favorable for the subsequent numerical optimization.

Accordingly, a geometric feasibility enhancement strategy is introduced when constructing the complete initial solution. The initial buffering variables are first evaluated from the reference trajectory and the initial temporal mesh, and the corresponding buffered-footprint vertices are constructed using (10). The initial footprints are then checked interval by interval against the collision-avoidance constraints in (17)–(18).

For an interval that already satisfies the collision-avoidance constraints, its initial buffering variables are left unchanged. For an interval with an initial collision, the vehicle poses and the forward-reverse motion topology supplied by hybrid A* are retained, while only the corresponding buffering variables are progressively reduced. In this way, the buffered footprint is

locally contracted from its fully expanded theoretical footprint toward the physical vehicle footprint until the associated collision-avoidance constraints are satisfied.

This treatment exploits the collision-free physical footprint already provided by hybrid A* as a geometric baseline. Because the buffered footprint differs from the physical vehicle mainly through its additional expansion, reducing this expansion when an initial conflict with an obstacle occurs allows the optimizer to start from a more favorable geometric relationship with the obstacles without altering the basic motion topology of the reference path.

It should be emphasized that this local contraction is used only to improve the initial point of the NLP. Once the buffering variables are artificially reduced, their values generally no longer satisfy exactly the motion-dependent definitions in (6), and the contracted footprint can therefore no longer be interpreted as a complete envelope of the continuous swept region over the corresponding interval. In other words, the geometric feasibility enhancement is not intended to construct an initial point that strictly satisfies all NLP constraints. Instead, it preferentially reduces the residuals of the numerically difficult collision-avoidance constraints.

Once the formal optimization begins, the buffered-footprint definitions and validity conditions in (6)–(8), together with the collision-avoidance constraints in (17)–(18), remain unchanged. The optimizer must therefore restore consistency among the local motion, buffering variables, and collision constraints during the subsequent iterations. The proposed strategy affects only the NLP initialization and does not modify the feasible set or the safety requirements of the final optimization problem.

C. Temporal Feasibility Enhancement of the Initial Solution

In (19), each interval length Δt_k is a decision variable rather than a prescribed sampling period. This formulation provides additional freedom for minimum-time optimization, but it also directly couples the temporal discretization with the buffered-footprint model. According to (4), the local travel distance depends jointly on the vehicle velocity and the interval length. Therefore, the initial temporal discretization directly affects both the buffering variables in (6) and the geometric validity conditions in (8).

If an initial interval is too long, the local travel distance and heading variation over that interval may become too large, causing the local envelope conditions in (8) to be violated. Conversely, uniformly using very short intervals facilitates satisfaction of these conditions but substantially increases the number of collocation points and, consequently, the number of state, control, and auxiliary variables in the NLP. The initial temporal mesh is therefore constructed adaptively according to the local motion characteristics of the reference trajectory and the validity conditions of the buffered footprint.

First, the forward-reverse switching points are identified from the motion directions of the hybrid A* reference path, and the complete path is divided into local segments with consistent motion directions. For each segment, a velocity profile satisfying the velocity and acceleration bounds is constructed according to the path length, with zero velocity imposed at each switching point to ensure physical continuity between forward and reverse motions. The steering angle is then obtained from the local curvature of the reference path, while the corresponding acceleration and steering rate are calculated from the temporal variations of velocity and steering angle,

respectively. This produces a complete time-parameterized reference trajectory.

The resulting reference trajectory is then resampled using a sufficiently small uniform time step. This fine grid serves only as a candidate set for constructing the initial temporal mesh and is not used as the fixed discretization of the final NLP. The initial time, terminal time, and all forward-reverse switching times are retained as mandatory collocation points so that no optimization interval crosses a change in motion direction.

Starting from the current retained time t_k , subsequent candidate times on the fine grid are examined sequentially. For each candidate interval, its duration, curvature, and smoothed forward and reverse travel distances are evaluated from the corresponding reference state. The interval is then checked against the local validity conditions in (8), together with the prescribed maximum interval length $\Delta t_{\max}$. Once extending the interval to the next resampled point would violate any of these conditions, the last admissible candidate is retained as the next collocation point t_{k+1} , and the same procedure is restarted from that point.

To prevent the initial intervals from lying too close to the theoretical validity boundaries in (8), a tightening factor is further introduced during initialization,

$$0 < \eta < 1 \quad (27)$$

Using the smoothed travel distances $\tilde{s}_k^+$ and $\tilde{s}_k^-$ employed in the actual optimization, the local validity conditions in (8) are tightened to

$$|\kappa_k| \cdot (\tilde{s}_k^+ + \tilde{s}_k^-) \leq \eta \cdot \frac{\pi}{2}, \quad (28a)$$

$$\left(1 + \frac{L_B}{2} \cdot |\kappa_k|\right) \cdot \tan(|\kappa_k| \cdot \tilde{s}_k^+) \leq \eta \cdot L_R \cdot |\kappa_k|, \quad (28b)$$

$$|\kappa_k| \cdot L_F \cdot \tan(|\kappa_k| \cdot \tilde{s}_k^+) \leq \eta \cdot \left(1 + \frac{L_B}{2} \cdot |\kappa_k|\right), \quad (28c)$$

$$\left(1 + \frac{L_B}{2} \cdot |\kappa_k|\right) \cdot \tan(|\kappa_k| \cdot \tilde{s}_k^-) \leq \eta \cdot L_F \cdot |\kappa_k|, \quad (28d)$$

$$|\kappa_k| \cdot L_R \cdot \tan(|\kappa_k| \cdot \tilde{s}_k^-) \leq \eta \cdot \left(1 + \frac{L_B}{2} \cdot |\kappa_k|\right), \quad (28e)$$

The tightening factor η is used only for constructing the initial temporal mesh and does not replace the original validity conditions in (8) during the formal optimization. Decreasing η causes candidate intervals approaching the theoretical validity boundaries to be terminated earlier, resulting in shorter initial intervals and more collocation points. This provides greater local adjustment margin for the subsequent optimization of velocity, steering, and time variables, at the expense of a larger NLP. Hence, η regulates the tradeoff between initialization margin and optimization problem size.

APPENDIX

Appendixes, if needed, appear before the acknowledgment.